

Hydromorphone

(Dilaudid)

- I. **Classification** – Opioid Analgesics
- II. **Action(s)**
 - Binds to opiate receptors in the CNS
 - Alters the perception of and response to painful stimuli while producing generalized CNS depression
- III. **Indication(s)**
 - Severe pain
- IV. **Contraindications**
 - Known hypersensitivity
- V. **Precautions**
 - Hydromorphone is a CNS depressant – All patients require SPO2 and ETCO2 monitoring
 - Severe renal disease (consider decreasing dose by 50% in renal patients)
 - History of substance abuse (if alternative is available, otherwise must be given with expressed consent from patient, parent, or guardian, and only in the setting of condition that would be painful)
 - Elderly or debilitated patients (dosage reduction suggested)
- VI. **Drug interactions**
 - Buprenorphine may diminish the therapeutic effects of opioid agonists
 - CNS depressants may enhance the CNS depression of opioid agonists
 - Desmopressin – Opioid agonists may enhance the hyponatremic effect of Desmopressin
 - Diuretics: Opioid agonists may enhance the toxic/adverse effects of diuretics
 - Kratom mimics the effects of an opioid and effects may be enhanced
- VII. **Adult Dosing and Administration**
 - 0.5 mg IV/IO/IM Repeat in 0.5mg increments every 20 minutes until desired effect up to total 2 mg
 - Consider 50% dosing for elderly and renal disease patients
 - Monitor respiratory and ventilatory status utilizing continuous SpO2 and ETCO2
 - Call On-Call Medical Control Physician for additional doses
- VIII. **Pediatric Dosing and Administration**
 - Contact Medical Control Physician for dose if pain is not managed by other means
- IX. **Onset**
 - 5 minutes
- X. **Duration**
 - 3-4 hours
- XI. **Possible Adverse Effects**
 - Constipation
 - Opioid-induced neurotoxicity – hyperalgesia, delirium, hallucinations, myoclonus, allodynia, tremor, seizures
 - Opioid induced pruritic
 - Opioid induced respiratory depression

XII. Other Notes

- Administer slowly! Rapid administration may lead to increased respiratory depression, respiratory muscle rigidity, hypotension, and circulatory change
- Assess level of consciousness, pulse, and respirations before and periodically after administration
- Monitor ETCO₂ after administration